

chapter 7

design rules

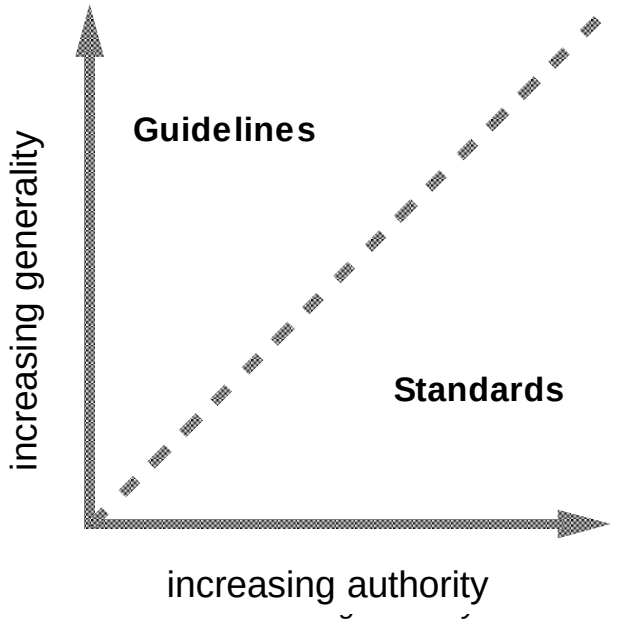
design rules

Designing for maximum usability
– the goal of interaction design

- Principles of usability
  – general understanding
- Standards and guidelines
  – direction for design
- Design patterns
  – capture and reuse design knowledge

types of design rules

- principles
  - abstract design rules
  - low authority
  - high generality
- standards
  - **specific** design rules
  - high authority
  - limited application
- guidelines
  - lower authority
  - more general application
  - Use clear labels for buttons



Principles to support usability

Learnability

the ease with which new users can begin effective interaction and achieve maximal performance

Flexibility

the multiplicity of ways the user and system exchange information

Robustness

the level of support provided the user in determining successful achievement and assessment of goal-directed behaviour

Principles of learnability

Predictability

- determining effect of future actions based on past interaction history
- operation visibility

Principles of learnability (ctd)

Familiar
- how prior knowledge applies to new system
- guessability; affordance

Generalizability
- extending specific interaction knowledge to new situations.

Consistency
- likeness in input/output behaviour arising from similar situations or task objectives.
- Enter /shift

Principles of flexibility

Dialogue initiative
- freedom from system imposed constraints on input dialogue
- system vs. user pre-emptiveness

Multithreading
- ability of system to support user interaction for more than one task at a time

Task migratability
- passing responsibility for task execution between user and system.

Principles of flexibility (ctd)

- allowing equivalent values of input and output to be replaced for each other
- representation multiplicity; equal opportunity

Customizability

- modifiability of the user interface by user (adaptability) or system (adaptivity).
- Change font.

Principles of robustness

Observability

- ability of user to evaluate the internal state of the system from its perceivable representation
- browsability; defaults; reachability; persistence; operation visibility.

Recoverability

- ability of user to take corrective action once an error has been recognized
- reachability; forward/backward recovery; commensurate effort

Principles of robustness (ctd)

Responsiveness

- how the user perceives the rate of communication with the system

Task conformance

- degree to which system services support all of the user's tasks
- task completeness

Standards

- set by national or international bodies to ensure compliance by a large community of designers standards require sound underlying theory and slowly changing technology.
- hardware standards more common than software high authority and low level of detail
- ISO 9241 defines usability as effectiveness, efficiency and satisfaction with which users accomplish tasks

Guidelines

- more suggestive and general in many textbooks and reports full of guidelines
- **Types:**
- abstract guidelines (principles) applicable during early life cycle activities
- detailed guidelines (style guides) applicable during later life cycle activities
- understanding justification for guidelines aids in resolving conflicts

Golden rules and heuristics

- "Broad brush" design rules
- Useful check list for good design
- Better design using these than using nothing!
- Different collections e.g.
  - Nielsen's 10 Heuristics (see Chapter 9)
  - Shneiderman's 8 Golden Rules
  - Norman's 7 Principles

Shneiderman's 8 Golden Rules

1. *Strive for consistency*
2. *Enable frequent users to use shortcuts*
3. *Offer informative feedback*
4. *Design dialogs to yield closure*
5. *Offer error prevention and simple error handling*
6. *Permit easy reversal of actions*
7. *Support internal locus of control*
8. *Reduce short-term memory load*

Norman's 7 Principles

1. *Use both knowledge in the world and knowledge in the head.*
2. *Simplify the structure of tasks.*
3. *Make things visible: bridge the gulfs of Execution and Evaluation.*
4. *Get the mappings right.*
5. *Exploit the power of constraints, both natural and artificial.*
6. *Design for error.*
7. *When all else fails, standardize.*

Summary

Principles for usability

- repeatable design for usability relies on maximizing benefit of one good design by abstracting out the general properties which can direct purposeful design
- The success of designing for usability requires both creative insight (new paradigms) and purposeful principled practice

Using design rules

- standards and guidelines to direct design activity